

29 Plant Diversity I: How Plants Colonized Land 618

CONCEPT 29.1 Plants evolved from green algae 619

- Evidence of Algal Ancestry 619
- Adaptations Enabling the Move to Land 619
- Derived Traits of Plants 621
- The Origin and Diversification of Plants 621

CONCEPT 29.2 Mosses and other nonvascular plants have life cycles dominated by gametophytes 623

- Bryophyte Gametophytes 624
- Bryophyte Sporophytes 625
- The Ecological and Economic Importance of Mosses 627

CONCEPT 29.3 Ferns and other seedless vascular plants were the first plants to grow tall 629

- Origins and Traits of Vascular Plants 629
- Classification of Seedless Vascular Plants 631
- The Significance of Seedless Vascular Plants 633

30 Plant Diversity II: The Evolution of Seed Plants 636

CONCEPT 30.1 Seeds and pollen grains are key adaptations for life on land 637

- Advantages of Reduced Gametophytes 637
- Heterospory: The Rule Among Seed Plants 638
- Ovules and Production of Eggs 638
- Pollen and Production of Sperm 638
- The Evolutionary Advantage of Seeds 639

CONCEPT 30.2 Gymnosperms bear “naked” seeds, typically on cones 640

- The Life Cycle of a Pine 640
- Early Seed Plants and the Rise of Gymnosperms 641
- Gymnosperm Diversity 641

CONCEPT 30.3 The reproductive adaptations of angiosperms include flowers and fruits 644

- Characteristics of Angiosperms 644
- Angiosperm Evolution 647
- Angiosperm Diversity 649

CONCEPT 30.4 Human welfare depends on seed plants 651

- Products from Seed Plants 651
- Threats to Plant Diversity 651



31 Fungi 654

CONCEPT 31.1 Fungi are heterotrophs that feed by absorption 655

- Nutrition and Ecology 655
- Body Structure 655
- Specialized Hyphae in Mycorrhizal Fungi 656

CONCEPT 31.2 Fungi produce spores through sexual or asexual life cycles 657

- Sexual Reproduction 658
- Asexual Reproduction 658



CONCEPT 31.3 The ancestor of fungi was an aquatic, single-celled, flagellated protist 659

- The Origin of Fungi 659
- The Move to Land 660

CONCEPT 31.4 Fungi have radiated into a diverse set of lineages 660

- Cryptomycetes and Microsporidians 661
- Zoopagomycetes 662
- Mucoromycetes 663
- Ascomycetes 663
- Basidiomycetes 665

CONCEPT 31.5 Fungi play key roles in nutrient cycling, ecological interactions, and human welfare 667

- Fungi as Decomposers 667
- Fungi as Mutualists 667
- Practical Uses of Fungi 670

32 An Overview of Animal Diversity 673

CONCEPT 32.1 Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers 674

- Nutritional Mode 674
- Cell Structure and Specialization 674
- Reproduction and Development 674

CONCEPT 32.2 The history of animals spans more than half a billion years 675

- Steps in the Origin of Multicellular Animals 675
- Neoproterozoic Era (1 Billion–541 Million Years Ago) 676
- Paleozoic Era (541–252 Million Years Ago) 677
- Mesozoic Era (252–66 Million Years Ago) 679
- Cenozoic Era (66 Million Years Ago to the Present) 679

CONCEPT 32.3 Animals can be characterized by body plans 679

- Symmetry 679
- Tissues 679
- Body Cavities 680
- Protostome and Deuterostome Development 681

CONCEPT 32.4 Views of animal phylogeny continue to be shaped by new molecular and morphological data 682

- The Diversification of Animals 682
- Future Directions in Animal Systematics 684